Iris Phenolic Liners
Continuous Heating to Model Exhaustion

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How long until the gas reaches the casing?

For the sustained nominal load, the independent moving-wall calculation exposes the casing after about 104–111 s in the sustainer and 138–148 s in the booster. The clock starts when hot gas first reaches bare phenolic. Burnthrough occurs when the moving char surface reaches the casing.

Motor	Liner [mm]	Radial [s]	ANSYS [s]	Earlier radial casing event
Sustainer Al	4.76	110.5	110.6	All Al liquid at 99.1 s
Sustainer CF	4.76	104.3	104.3	Casing at 1000 °C at 95.1 s
Booster Al	6.35	147.6	147.7	All Al liquid at 139.4 s
Booster CF	6.35	137.9	138.0	Casing at 1000 °C at 129.7 s

The calculation applies 2504 K gas continuously at 800 psi with an initial film coefficient of 1000 W/(m^2 K), gas blowing and zero hot-side radiation. The Al comparison follows pure-metal fusion and retains the molten metal in place. The CF comparison follows oxygen-free T700/M21 resin decomposition and holds the residue in place. The study has no measured Iris wall heat flux or fired-liner recession. All four times therefore describe the stated thermal scenarios. The casing can lose pressure capacity while phenolic remains.

What most affects the estimate. Halving and doubling the nominal film coefficient gives 57–207 s for the sustainers and 75–276 s for the boosters with the evolving casings. Section 12 gives all twelve times and the heat flux entering the liner. With nominal heating, changing only the energy required to remove a kilogram of char gives 72–185 s for the aluminium sustainer (Section 19). A measured firing heat-flux history and liner recession would constrain the physical estimate.

Verification. Each native calculation started with both bodies at 22 °C and continued to zero liner thickness (Section 18). The prescribed gas is 2231 °C. The highest sampled exposed solid surface temperature in the four native runs was about 1077 °C (1350 K). The two independent methods at matched fine resolution gave event times within 0.008 s of each other. Their last integration intervals are disjoint (Section 21). Table values support a seconds-scale comparison for this loading. The actual time for the firing motor depends on the delivered heat and the response of the manufactured wall.

1 Baseline comparison with fixed casing properties

With the original casing properties, the moving-wall calculation gives about 104–107 s for the sustainer and 138–142 s for the booster under the nominal continuous load. It advances the gas-facing radius until it reaches the casing. Char can remain after the virgin resin has decomposed, so conversion and casing exposure occur at different times. This is an independent finite-volume calculation. ANSYS full-exposure results with an evolving casing appear in Section 18.

Case	Initial liner [mm]	Nominal event [s]	Property flag [s]	Film scenarios [s]
Sustainer Al	4.7625	106.8	49.0	58.3–200.1
Sustainer CF	4.7625	103.9	20.5	56.5–194.9
Booster Al	6.3500	142.4	82.0	77.8–266.3
Booster CF	6.3500	137.7	35.1	74.7–258.3

Table 1: Nominal and film endpoints use 640 liner and 160 casing cells with a 0.02 s step ceiling. Film scenarios change the uncorrected coefficient from 2000 to 500 W/(m^2 K), respectively. The other inputs retain nominal values. The property flag in this table belongs to the nominal case.

All events occur after the assumed intact casing exceeds its provisional property range. The nominal gas temperature is 2504 K, nominal chamber pressure is 800 psi, and the nominal uncorrected film coefficient is 1000 W/(m^2 K). The calculation exposes the liner at ignition and sustains heating until full local loss. Zero hot-side radiation is assumed at nominal conditions. The outside sees a 22 °C environment, an 8 W/(m^2 K) film and emissivity 0.25. The extent of actual in-motor exposure is unknown.

The largest one-factor changes in this study come from wall heat transfer and the assumed energy cost of char removal. The times in the last column compare two specified heat-transfer scenarios. They do not quantify firing-to-firing variability or a safe firing interval. Firing heat flux versus time and recession measurements on the manufactured liner would directly constrain the two inputs driving this spread. Section 6 shows how accounting for aluminium melting changes the liner times and when the backing melts.

2 Remaining thickness and casing heat-up

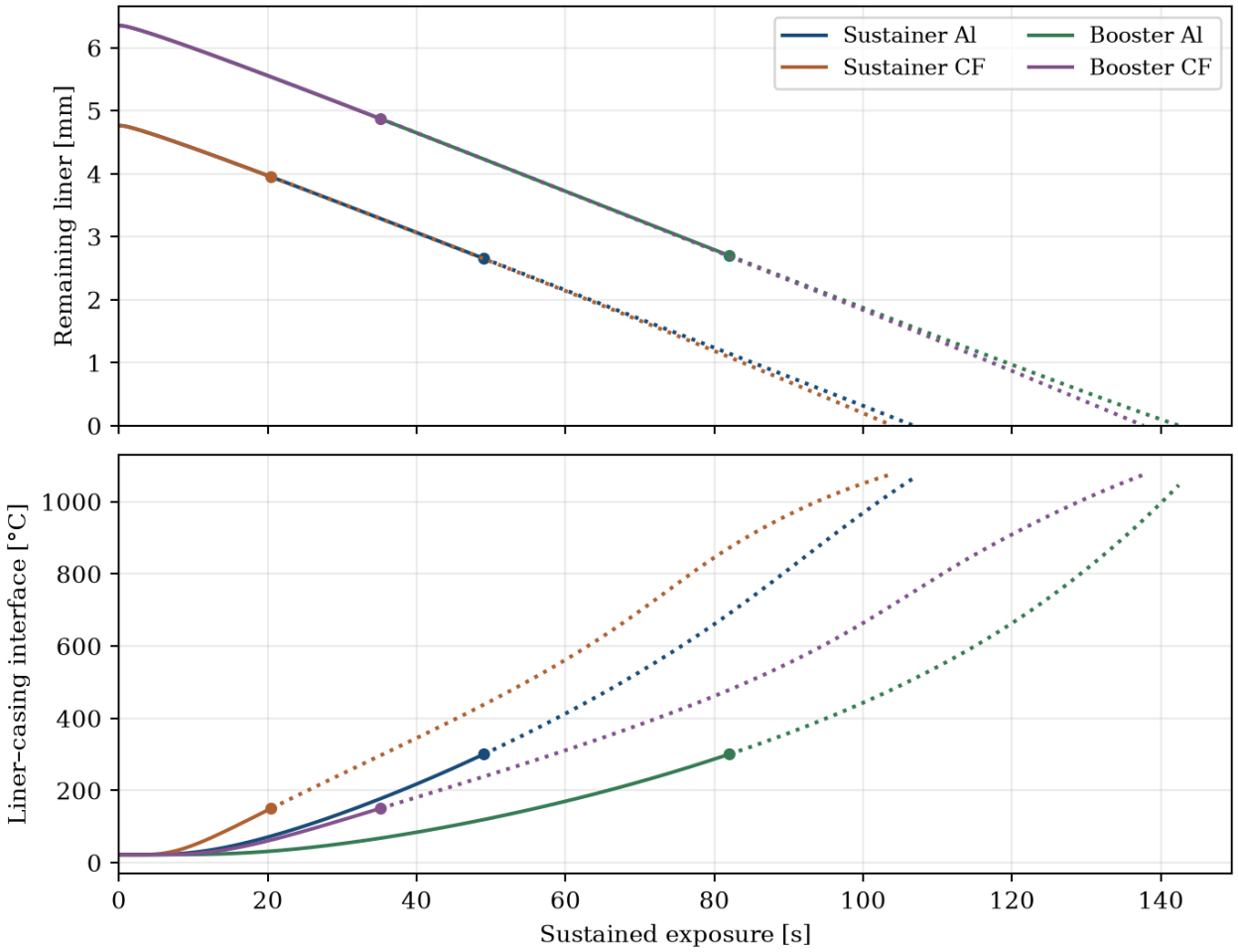


Figure 1: Moving-wall calculation at the finest nominal setting. Dots mark the first 150 °C CF or 300 °C Al property flag. Dotted lines continue with the inherited backing properties. The horizontal axis measures continuous exposure of the local wall.

The two casings give similar early surface recession because the heat initially penetrates only partway through the liner. The model separates them more clearly later as the casing absorbs heat. Each property flag denotes the limit adopted for using those casing data. It does not identify loss of pressure containment.

3 Temperatures, mass and energy in the sustainer Al calculation

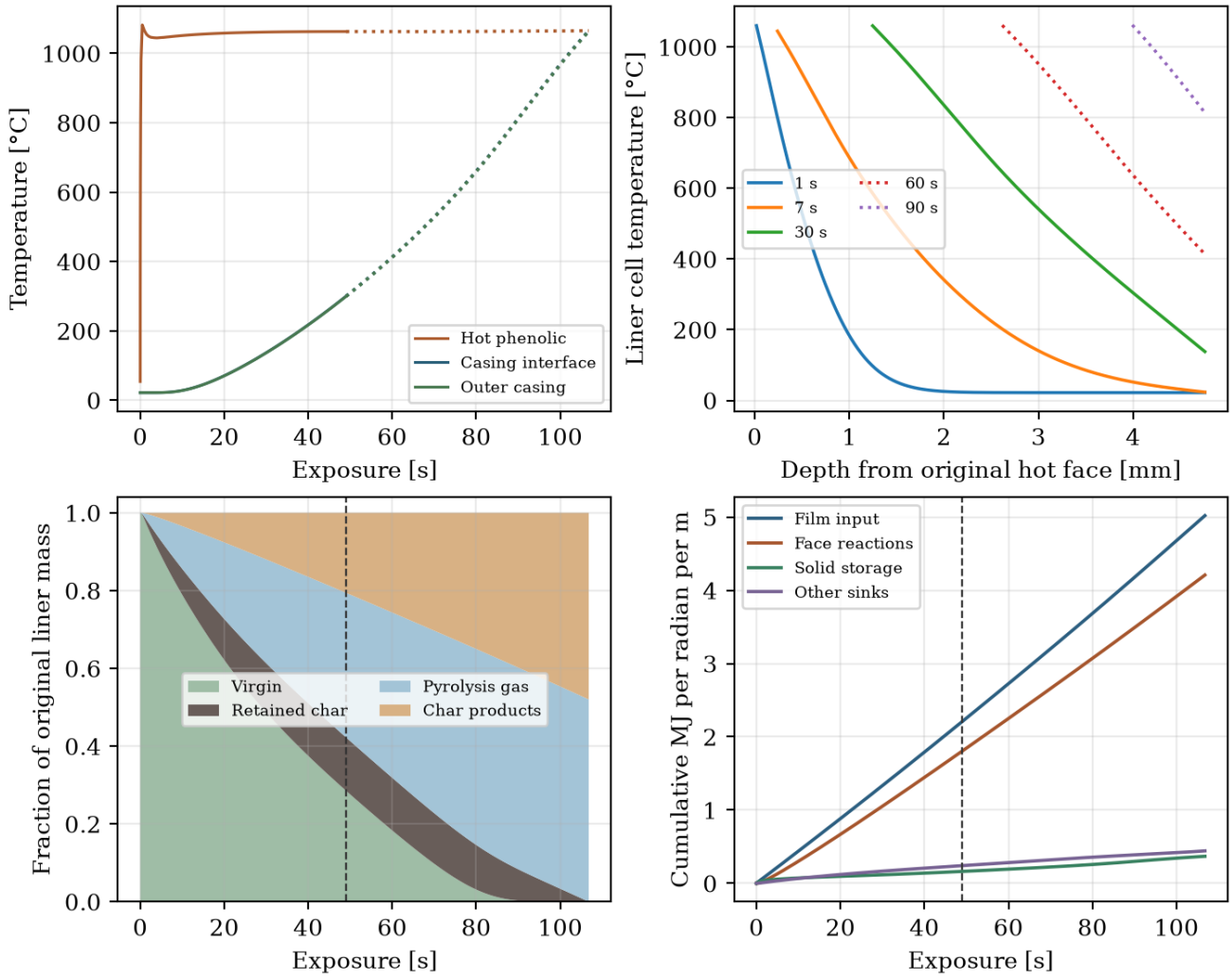


Figure 2: Upper panels show the three wall temperatures and the spatial temperature profiles of the surviving liner. Profiles start at the first cell centre. Lower panels account for the original liner mass and cumulative energy. Dashed vertical lines mark the casing property flag; subsequent values use surrogate continuation.

At 7 s the coupled wall gives 1048 °C at the hot phenolic face and 0.235 mm removed. At the same exposure time the accepted earlier native 320-row sustainer Al calculation gives 1444 °C at its live face and 0.045 mm of removed mesh. The gas is 2230.85 °C in both cases. The native calculation uses finite-row removal and its inherited reaction and energy equations. This difference therefore includes changes to the model as well as discretization; matching one final event time could conceal a substantial difference in surface behaviour.

At nominal exhaustion the surface reaction terms account for 83.8 percent of integrated film input in this case. The provisional 35 MJ/kg char cost therefore warrants a direct sensitivity check. Swept sensible heat appears on both sides of the interface accounting and cancels from the whole-wall ledger.

4 Numerical checks and open questions

Twenty nominal calculations isolate the time step at 160 cells, double the liner and casing radial counts twice at the same step, then halve the step on the 640/160 mesh. The last two comparisons below use 320 to 640 liner cells at 0.04 s and 0.04 to 0.02 s at 640 cells.

	Event shift [s]		Thickness	Mesh ΔT_s [K]	
Case	Mesh	Time	diff. [mm]	All samples	From 0.5 s
Sustainer Al	-0.0019	+0.0009	0.00016	30.3	2.4
Sustainer CF	-0.0018	-0.0003	0.00016	30.3	2.4
Booster Al	-0.0032	+0.0029	0.00024	39.3	4.7
Booster CF	-0.0030	+0.0000	0.00024	39.3	4.7

Table 2: Thickness difference is the largest across both comparisons at shared times. The temperature columns give the greatest hot-face difference between the two spatial resolutions, including or excluding the initial transient.

All eight finest pairs meet the 5 percent event-time target and the remaining-thickness target. Their final accepted time intervals overlap. Three coarser pairs have disjoint intervals. A load applied instantaneously to a massless reconstructed face gives a mesh-sensitive initial temperature. The 640-cell result still requires a separate comparison with ANSYS under matching equations, even where independent refinement differences are small.

The coupled implementation closes each cell energy equation before moving the front. The accepted step limits conversion growth to 0.05 and surface travel to 40 percent of the first-cell width while multiple cells remain. A remnant below half the original pitch merges conservatively with its neighbour. The last event is localized by solving for the step duration that brings the radius to the interface. A manufactured isothermal char front reproduces its analytical time to within 10^{-8} s at two resolutions. A nonuniform reacting coupon, interrupted after mesh mergers and a property flag, gives a byte-identical final state and history on restart. All twenty nominal integrations close the cumulative open-system energy and mass ledgers to relative residuals below 10^{-9} and 10^{-11}, respectively.

5 One-factor sensitivity to assumed inputs

Forty-two additional calculations each change one named input or one backing material tuple at 320/80 cells and a 0.04 s step ceiling. The columns show paired event times in the same order as their listed inputs, in seconds.

Case	Pressure 600 / 1000 psi	Radiation 0.1 / 0.3	Removal cost 17.5 / 70 MJ/kg	Gas excess 0.9 / 1.1
Sustainer Al	109 / 105	92 / 73	68 / 181	127 / 92
Sustainer CF	106 / 103	90 / 71	65 / 179	124 / 89
Booster Al	145 / 140	123 / 97	91 / 242	170 / 123
Booster CF	140 / 136	119 / 93	85 / 238	164 / 118

Radiation entries specify effective hot-side emissivity at the inherited gas radiative temperature. Gas excess scales the 2208.85 K difference between nominal gas and the initial wall, with convection coefficient and all other parameters fixed. The gas probes measure local sensitivity. Pressure enters the empirical consumption law; the film coefficient stays independent.

For each case, doubling and halving the film coefficient produce respectively the earliest and latest event in this screen. Both endpoints were rerun at 640/160 and 0.04 s, then 0.02 s (sixteen checks). The largest individual time shift is 0.007 s; 15 of sixteen pairs have overlapping final intervals. All sixteen satisfy the event-change and shared-thickness criteria. Individual endpoint intervals and the material crossings are recorded with the cases. An alternative inherited carbon-composite property tuple shifts the CF event by -0.84 s for the sustainer and -1.02 s for the booster. The resin, layup, bond and grade used on the actual cases remain unspecified.

Sections 6 and 7 examine aluminium fusion and CF resin decomposition with separate material surrogates.

6 Melting in a retained aluminium backing

The original aluminium cases use constant specific heat. The alternative stores the heat of fusion and allows the backing to become liquid while retaining its initial mass, shape, radial conductivity of 167 W/(m K) and density of 2700 kg/m^3. Gas loading, liner chemistry and dimensions match the nominal cases in Table 1. The extra heat absorbed by the casing delays liner exhaustion by 3.7 s in the sustainer and 5.2 s in the booster.

Case	Property flag [s]	First melt [s]	All cells liquid [s]	Liner gone [s]	Liner at full melt [mm]
Sustainer	50.7	84.1	99.1	110.5	0.43
Booster	84.4	124.4	139.4	147.6	0.23

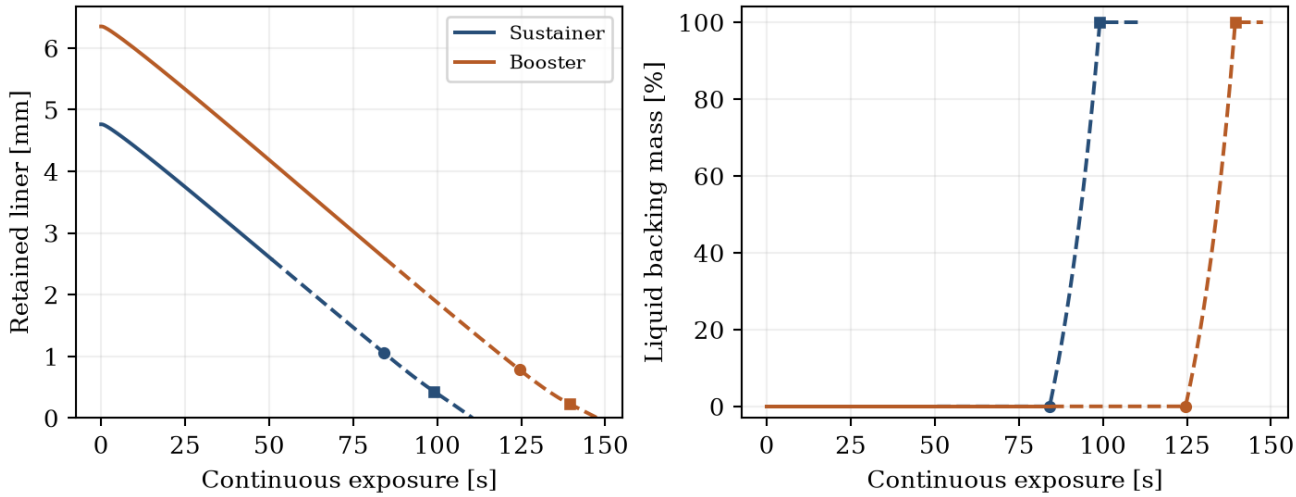


Figure 3: Solid curves end at the inherited 300 °C property flag. Dashes continue with the retained pure-Al surrogate. Circles mark the first cell with liquid and squares mark the first instant when all casing cells are liquid.

At complete modelled melting the liner still covers the case in both geometries. Following the liner further assumes the liquid stays in place. Pressure containment, flow of molten metal, alloy solidus and local wall defects need separate evidence.

The caloric law uses the crystal and liquid rows on the same NIST-JANAF enthalpy reference. At 933.45 K the 10.711 kJ/mol jump corresponds to 397.0 kJ/kg. The solver joins the two sensible branches across a narrow numerical melting interval, then recovers liquid fraction from the sharp equilibrium enthalpy. Ten integrations separate mesh, time-step and interval-width changes. The reported runs use 640/160 cells, a 0.02 s step ceiling and a 0.5 K interval. Across the eight single-parameter comparisons the greatest liner-event shift was 0.007 s. All eight satisfy the thickness and final-interval overlap checks. An externally heated char/Al coupon agrees with an independent sharp-transition enthalpy ODE to 0.25 K at the finest setting.

7 Resin reaction on the CF temperature histories

A separate calculation applies the two competing T700/M21 reactions measured under nitrogen to the liner-side and outside temperatures of the published CF cases. The heat histories are prescribed. Heat absorption, gas transport and the changing composite conductivity would have to feed back into the wall to revise a CF liner-exhaustion time.

Let $u = 1 - a_1 - a_2$ denote the fraction of the shared reactant still available. With T in kelvin and $k_j = A_j \exp[-E_j/(RT)]$ in s^{-1}, the rates are

$$\dot{a}_1 = k_1 u^{1.1125}, \qquad \dot{a}_2 = k_2 u^{2.0825}(1 + K_{\mathrm{cat}} a_2).$$

The frequency factors are 7.2193 and 5.3309×10^8 s^{-1}, and the activation energies are 58.7107 and 146.674 kJ/mol. The table uses $K_{\mathrm{cat}} = 7.4886$ from Table 5 of the experimental paper. The model equations and competitive mass-loss weights are given in its companion paper. Tranchard et al., Part I; Part II.

		Time to reactant conversion [s]			At 50%
Case	Casing location	10%	50%	90%	[°C]
Sustainer	Liner side	52.4	56.6	61.4	522
Sustainer	Outside	71.2	75.7	80.7	520
Booster	Liner side	80.5	85.9	91.8	514
Booster	Outside	117.5	123.2	129.8	511

Here 50% means half of the common reactive pool has advanced by either pathway. Reconstructing mass with the paper's density change and pathway weights gives about 12.3% loss of the initial composite mass at that point. The printed $\log_{10} K_{\mathrm{cat}} = -0.8744$ in Part I, Table 3 implies a smaller coefficient. Using that interpretation delays the four 50% times by 3.0–4.1 s. The calculations stop at 99.9% common conversion, 1000 °C, or the end of the supplied temperature history, whichever occurs first.

The local rates of temperature rise at 50% conversion are 492–697 °C/min, above the reported nitrogen tests at 5–100 °C/min. The Iris casing resin and layup are unspecified. These times identify when a reacting backing can affect the heat balance. The following section couples heat storage, radial conductivity, decomposition energy and escaping gas to both motor geometries.

8 CF casing heat, gas loss and liner exposure

The backing now feeds its changing mass, heat storage and through-thickness conductivity back into the thermal solution. Gas at 2504 K continues to heat the liner. The resin's two competing rates follow Section 7. Each casing cell releases gas at its own temperature into an assumed separate vent. This leaves the liner-side blowing calculation dependent on phenolic gas alone. The radial model holds the decomposed casing in its original shape.

Motor	Half reactive pool [s] Inner cell	Outer cell	Casing at 1000 °C [s]	Liner then [mm]	Liner gone [s]
Sustainer	60.7	82.4	95.1	0.45	104.3
Booster	91.4	135.3	129.7	0.40	137.9

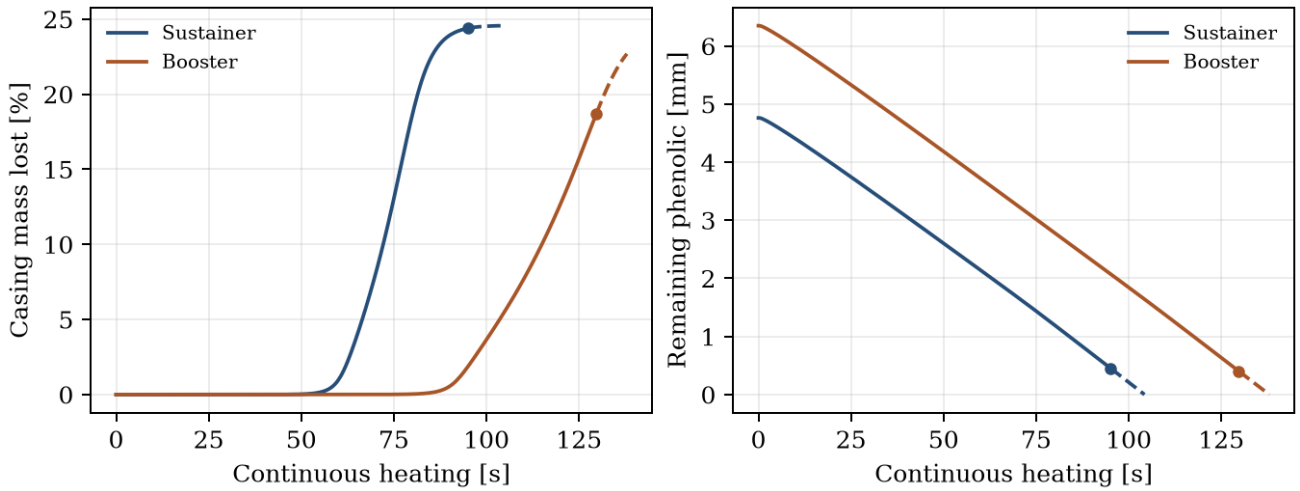


Figure 4: Gas release as a percentage of initial casing mass and remaining liner for the nominal reacting CF cases. Dots show the first backing temperature of 1000 °C. Dashed segments hold the upper-end residue and gas properties beyond this limit. Virgin-property fits reach their own limits at lower temperatures.

At liner exhaustion the surrogate releases 24.6% of the sustainer casing mass and 22.7% of the booster casing mass. The detailed backing changes the nominal liner time by +0.32 and +0.28 s respectively relative to the inert CF baseline. Changing the catalyst coefficient alone shifts these results by +0.084 and +0.122 s. The local resin, bond and layup in Iris are unknown. No structural failure time is inferred.

Local thermodynamics. For competing extents a_1, a_2, define $x = 0.554a_1 + 0.9479a_2$, $\rho = 1575 - 410x$ in kg/m^3 and $\Delta g_i = 410\chi_i \Delta a_i V$ in kg. The remaining cell mass is $m = \rho V$. With sensible enthalpies measured from 22 °C, the reaction substep closes

$$(mh_s)_{j+1} - (mh_s)_j = \sum_i Q_i \Delta g_i - h_g(T_{1/2}) \sum_i \Delta g_i.$$

Here $Q_1 = +259.54$ and $Q_2 = -152.22$ kJ per kg of generated gas, following the density-loss source in Tranchard et al., Part II. Positive Q_i adds heat. The model mixes virgin and degraded sensible enthalpies by x. Differentiating this stored energy includes its composition dependence.

9 CF transport properties and numerical checks

The model mixes normal conductivity and specific heat between virgin and degraded states using x. For temperature T in °C the virgin fits are $c_v = 687.31+2.8773T$ and $k_v = 0.61391+0.0011113T$. The corresponding upper fit limits are 300 and 150 °C. The frozen endpoint value is used outside either fit. The degraded and gas fits below use a 20–1000 °C interval, also with endpoint continuation:

$$
\begin{aligned}
c_d &= 662.53 + 2.599T - 2.0761\,10^{-3}T^2 + 5.1327\,10^{-7}T^3,\\
k_d &= 0.077211 + 2.3397\,10^{-4}T - 4.1903\,10^{-7}T^2 + 7.4228\,10^{-10}T^3,\\
c_g &= 1256.6 + 1.0610T - 9.2485\,10^{-4}T^2 + 3.5977\,10^{-7}T^3.
\end{aligned}
$$

Specific heats use J/(kg K) and conductivities W/(m K). Integrating each bounded heat-capacity function gives its sensible enthalpy. Local radial conductances include the logarithmic resistance of both adjacent annular half-cells. The thermal substep iterates conductivity with temperature. The adiabatic reaction substep advances both pathways against a common reactant pool and solves for temperature from the open-system energy equation. A trial repeats at a smaller time step if the reaction fails to converge or if any cell's total channel advance exceeds 0.04. The backing vent removes gas enthalpy once at generation. Gas pressure, internal transport work, swelling and delamination are outside this closure. Material measurements, Part I; thermal model, Part II.

Checks. A two-cell coupon with a constant outside heat input compares this split solver against an independent temperature-form ODE. Over 28 s the finest 0.01 s step gives at most 0.032 K temperature error and 0.0011 absolute error in either pathway. Both nominal motor pairs pass their mesh and time checks on the event brackets and remaining thickness. The finest comparisons use 640/160 liner/casing cells and a 0.02 s ceiling. Across the eight runs, each recorded profile independently reconstructs casing mass and total sensible enthalpy. An interrupted warm coupon recovered both reaction channels and produced identical final files after restart. These checks test the specified equations. Quantitative prediction for Iris still depends on a measured heat load, recession calibration and material-specific validation.

10 A second calculation from ignition to liner exhaustion

A second implementation places temperature, virgin mass and char mass at the nodes of a moving finite-element mesh. The preceding finite-volume calculation stores its material state at cell centres and solves for a separate surface temperature. Both follow the same assumed heat exposure and backing properties. The finite-element results below use the radial reduction of the sector equations, evaluated on the Mac. Sections 15–17 give the separate Windows MAPDL checks.

	Finite volume	Finite element [s]		
Case	640/160 [s]	80/20, 0.04 s	160/40, 0.04 s	160/40, 0.02 s
Sustainer Al	110.47	110.58	110.58	110.53
Sustainer CF	104.25	104.33	104.23	104.29
Booster Al	147.61	147.75	147.74	147.68
Booster CF	137.95	138.04	138.03	137.99

The mesh pairs count liner and casing intervals. The solver shortens steps at rapid reaction, fusion, front motion and output stations. In these twelve runs, doubling the radial resolution changed the endpoint by at most 0.10 s. Halving the time step changed it by at most 0.061 s. The largest difference between the finest two methods was 0.073 s.

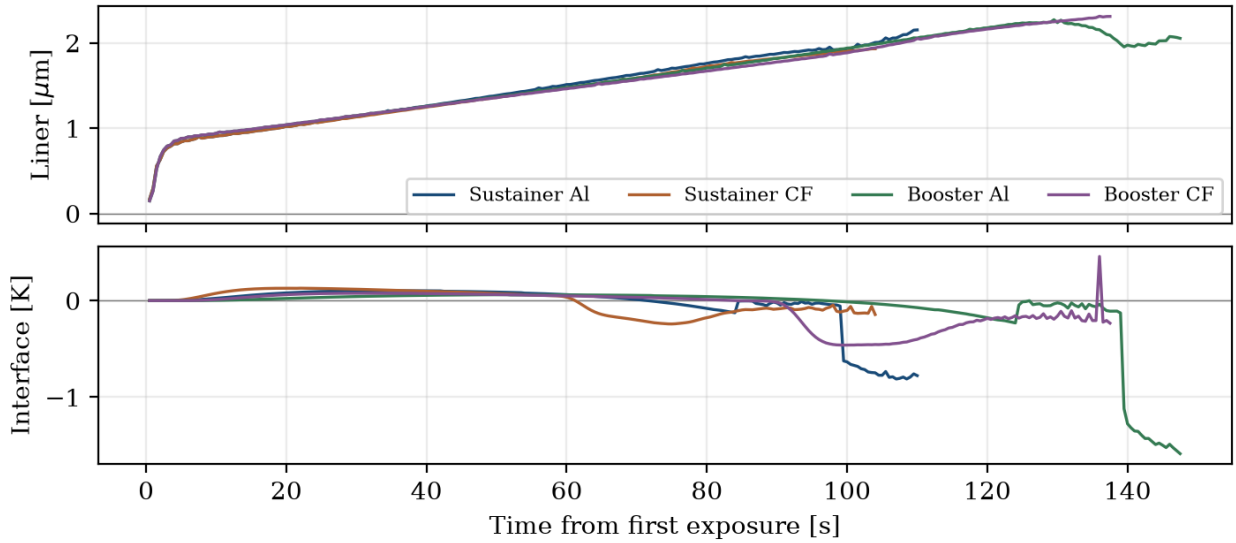


Figure 5: Finite element minus finite volume at common exposure times, sampled every half second before either run ends. The upper panel shows remaining liner thickness and the lower panel shows liner–casing interface temperature.

The largest sampled thickness difference is 2.3 μm and the largest interface difference is 1.59 K. All four cases pass their baseline casing property limits. Both aluminium walls are wholly liquid in the retained-shape model before liner exhaustion. Both CF walls pass 1000 °C beforehand. An uncertainty in the imposed heat transfer affects both numerical methods in the same direction. Their agreement gives no independent calibration of the actual firing heat or reaction rate.

11 Thermal step and recession in the moving finite-element sector

For a radial interval with boundaries a, b, the row-summed linear brick allocates volume per radian and per axial metre $(b-a)(2a+b)/6$ to its inner node. The outer node receives the remainder of $(b^2-a^2)/2$. The equivalent dual-volume boundary is $r_d^2 = a^2 + (b-a)(2a+b)/3$. These volumes carry virgin mass m_v, char mass m_c and sensible enthalpy $m_v h_v + m_c h_c$. At the liner–casing interface both material contributions enter one temperature equation. With straight sector faces of angular width δ, radial conductance and the backward-Euler transport equation are

$$G_e = \frac{k_e (a+b)}{2(b-a)\cos^2(\delta/2)}, \qquad (C/\Delta t + K + H)T^* = CT^n/\Delta t + f.$$

C is the diagonal secant heat capacity from the change of sensible enthalpy at fixed composition. It includes heat absorbed in aluminium fusion. H adds the hot and ambient film terms. The ambient coefficient includes radiation linearized at the current iterate. f includes the gas and ambient temperature terms, prescribed incident radiation and the surface energy consumed in forming and heating the products. The local gas film uses the previous bulk gas flow and the current surface product flow. Its reference gas temperature is 2504 K. T^* is the transported field before local reaction.

At the exposed node the reference char kinetics are multiplied by the fraction $m_c/(m_c + Y m_v)$, where $Y = 600/1250$ is the inherited char yield. For surface flux j and starting radius r_0, the step ends at r_1 satisfying

$$\int_{r_0}^{r_1} (\rho_c + Y\rho_v)\, r\, dr = \Delta t\; j\; \frac{r_0 + r_1}{2\cos(\delta/2)}.$$

Swept virgin material releases gas and supplies its char yield for consumption. The surface ledger includes the sensible enthalpy swept out, the sensible enthalpy of the exiting products, the heat of pyrolysis and the effective char-removal cost. The retained mass and sensible enthalpy are distributed over overlapping dual volumes. The code solves the temperature from enthalpy again after bulk pyrolysis and after equilibration at the shared interface. The time step is shortened to the instant the remaining convertible mass reaches zero. Restarting each short verification path reproduced its complete terminal state and ledgers. The largest cumulative energy discrepancy in the twelve full exposures was below 10^{-7} of incident film heat.

The prepared native implementation uses SOLID278 for conduction and MASS71 for the nodal thermal capacity beyond the solid element's unit volumetric capacity. The radial predictor sets material coefficients. MAPDL must then solve the temperatures of all sector nodes with the hot and backing faces free. The controller checks the returned field before updating chemistry or moving the surface. The first two short native jobs test this exchange and the saved fields before a whole exposure can be accepted.

An independent three-dimensional assembly on Windows parsed the actual generated elements, nodal capacities, initial conditions and surface loads. Eight Gauss points per brick integrated heat capacity and conduction. In addition to ignition, the checks cover the first cell merge, aluminium fusion or CF above 1000 °C, and the final live increment for each motor. The largest temperature difference from the radial reduction was below 10^{-9} K. These checks ran in the bundled Python interpreter. Running the same input through the native solver tests an additional software boundary.

12 How strongly does incident heating change the answer?

The following exposures start at 22 °C, with bare phenolic facing 2504 K gas at 800 psi. Each column changes the gas-side film coefficient before blowing, h_0, in W/(m^2 K). Wall temperature, gas release, char consumption and the casing response evolve together. The first three columns give the time to consume the original liner, including its char. The last three show when all aluminium cells become liquid or the hottest CF casing cell reaches 1000 °C. Times are in seconds.

	Liner exhausted			Casing event		
h_0	500	1000	2000	500	1000	2000
Sustainer Al	206.9	110.5	60.2	170.7	99.1	56.9
Sustainer CF	195.4	104.3	56.6	—	95.1	51.3
Booster Al	276.4	147.6	80.4	248.8	139.4	77.8
Booster CF	258.7	137.9	74.8	—	129.7	69.7

The dash means the CF surrogate stayed below 1000 °C until exhaustion. All four casings exceeded their original material-property flags at all three loads. The calculation retains the molten metal or decomposed composite in place.

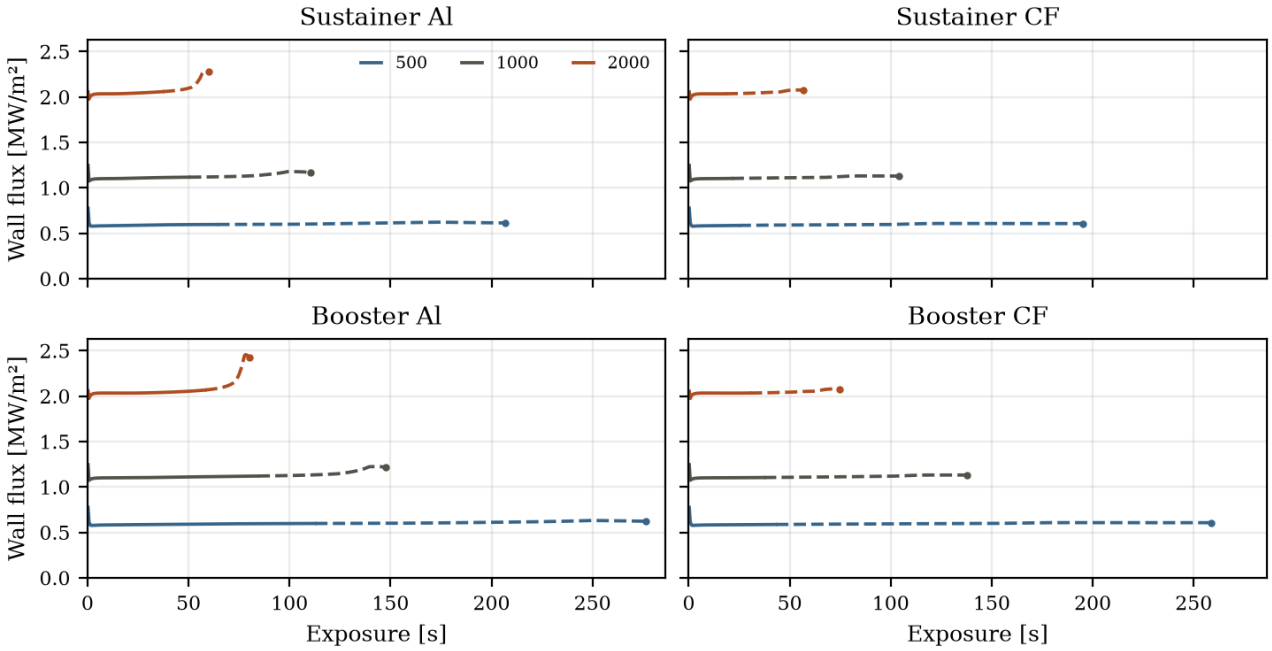


Figure 6: Modelled convective heat entering the receding surface over each half-second interval and the final partial interval. Colours identify h_0. Dashes start at the first 300 °C Al or 150 °C CF interface flag. The final dot marks the mean over the last interval before liner loss.

The film changes the rate at which energy enters the wall. Even with a fixed gas temperature and h_0, the driving temperature difference falls as the surface heats, and gas from the liner suppresses convection. The actual incident load requires a firing measurement or an independently validated gas-side calculation.

13 Heat delivered to the receding surface and numerical checks

For one radian of circumference and one metre of axial length, the exposed area equals its instantaneous radius r_s in SI units. Each saved interval provides its convective energy ΔE_f, including the blowing correction. The plotted flux divides this energy by the duration and average endpoint area. Because the surface moves monotonically outwards, the two endpoint areas also enclose the exact area–time integral. For the full exposure,

$$\frac{\sum \Delta E_f}{\sum r_{j+1}\Delta t_j} \le \overline{q''}_f = \frac{\int q''_f r_s \, dt}{\int r_s \, dt} \le \frac{\sum \Delta E_f}{\sum r_j \Delta t_j}.$$

The table gives the midpoint of these bounds in MW/m^2. Across all twelve histories their largest half-width is 1.022 kW/m^2. The last column gives cumulative film energy per original exposed area across the three loads.

	Mean flux [MW/m^2]			Energy per original area
h_0 [W/(m^2 K)]	500	1000	2000	[MJ/m^2]
Sustainer Al	0.603	1.126	2.074	131.9–132.6
Sustainer CF	0.598	1.114	2.045	122.5–123.4
Booster Al	0.603	1.128	2.079	174.8–175.6
Booster CF	0.597	1.113	2.043	160.2–161.6

Checking each heating level. For each of the eight new combinations, the finite-volume calculation was repeated with 320/80 and 640/160 liner/casing cells at a 0.04 s step ceiling, then with 640/160 cells at 0.02 s. An accepted pair must have overlapping event brackets, less than 5% endpoint change and matched thickness within the larger of 0.02 mm and 1% of original thickness. Booster Al at 2000 required additional time refinement. At 640/160 cells, the last two step ceilings were 0.005 and 0.0025 s. Their endpoints differed by 0.0013 s.

A second radial finite-element calculation used 160/40 cells and a 0.02 s ceiling from ignition in all eight cases. Across those cases the two methods differ by at most 0.127 s at exhaustion, 0.0066 mm of liner on common reporting times, and 10.93 K at the casing interface. For the finite-volume records the largest fractional mass imbalance is 2.2×10^{-12} and the largest heat imbalance relative to incident energy is 9.8×10^{-12}. These checks quantify numerical agreement for the stated closures.

What still controls the physical answer. In this calculation gas impinges directly on bare phenolic until its last material is removed. Actual ignition-to-exposure delay, burn duration, casing retention and circumferential hot spots can change the result. A local heat-flux history and pre/post-fire liner profiles at the same station would test both the imposed load and the predicted loss. The table on the preceding page gives a sensitivity to load, with no assigned probability.

14 Why the model gives minutes to liner loss

When the liner disappears, every kilogram of initial phenolic has passed through pyrolysis. In the specified chemistry, a fraction $y = 0.48$ of that mass becomes char and eventually leaves through the hot surface. For original hot radius r_i, thickness d and exposed area A_0, the original virgin mass per area follows from the volume of the annulus. The model's total chemical energy cost is then

$$\frac{m_0}{A_0} = \rho_v d \left(1 + \frac{d}{2r_i}\right), \qquad \frac{E_{\text{chem}}}{A_0} = \frac{m_0}{A_0}(L_p + yL_c).$$

The assumed costs are $L_p = 0.418$ MJ per kg of virgin phenolic and $L_c = 35$ MJ per kg of expelled char. They give 108.7 and 143.3 MJ/m^2 for the sustainer and booster, respectively. The calculation accounts separately for the sensible heat leaving with reaction products, the heat retained by the casing and the heat lost to ambient. For CF it also includes gas enthalpy and net resin reaction energy.

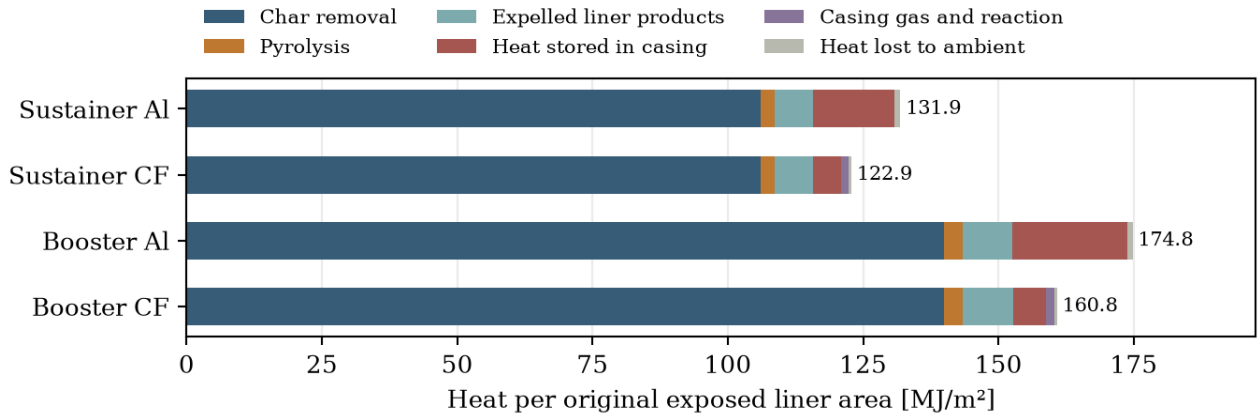


Figure 7: Energy delivered by the film through complete liner loss at nominal heating. The Al storage term includes fusion. The four bar totals use the same original-area basis as the chemical energy calculation. Both backing surrogates retain their shape.

For the sustainer Al example, the computed film delivers 131.9 MJ/m^2 at a time-average rate of 1.194 MW/m^2 on the original area. Their ratio is 110.5 s. The mean on the moving area in the preceding section is smaller because that area grows. This division explains the solved duration. Both the dose and the incident power come from the transient solution.

Across all twelve cases, 79.7–87.3% of incident energy pays the assumed char-removal cost. The two numerical methods agree on total dose to within 0.044%. A measured mass-loss and recession history under a known heat load would test the effective cost and the prescribed char yield. Their product sets the largest term in this estimate.

Gas and solid temperature. The prescribed gas temperature is 2504 K (2231 °C). Across the four motors the highest saved live hot-face temperatures are 975–988, 1082–1083 and 1229–1232 °C at the weaker, nominal and stronger film coefficients respectively. These are maxima of the saved surface samples. The burning gas, the exposed char and the casing interface have separate temperatures.

15 What the native ANSYS benchmarks verify

Both cylindrical faces are held at 22 °C. For a homogeneous hollow cylinder the first radial diffusion mode has the form $T(r,t) - 22 = \phi(r)\exp[-kt\lambda^2/(\rho c_p)]$, where ϕ vanishes at both boundaries. After one decay time the temperature rise is $1/e$ of its initial value everywhere. MAPDL advanced 256 substeps over 21.135 s on a sector of 256 bricks with 585 nodes. The largest nodal error was 0.079 K against a predeclared limit of 0.25 K. The root-mean-square error was 0.055 K.

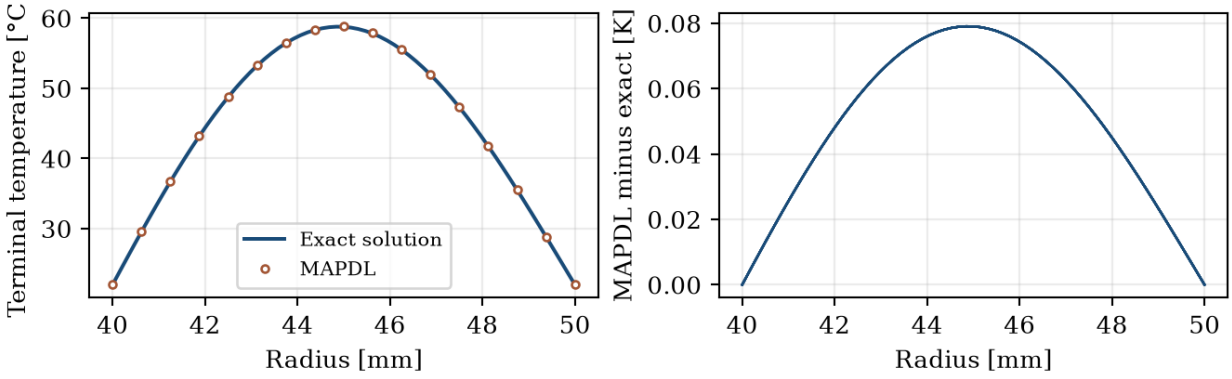


Figure 8: At left, the native solution and analytical profile after one decay time. At right, the signed error over all nine axial and angular node lines.

Receding char. A second benchmark uses a 0.6 mm pure-char wall with a prescribed logarithmic temperature profile and a corresponding heating law. Integrating the analytical recession speed gives the reference time. A first implementation distorted connected elements while moving nodes and also produced oscillating temperatures under a lagged surface heat sink. The revised calculation rebuilds each brick after moving all nodes and uses the local derivative of net incident flux to couple heating and char consumption. Narrower transverse dimensions and a shared final time interval control the aspect ratio as the remaining char becomes thin.

Radial cells	Steps	MAPDL [s]	Exact [s]	Error [%]
8	41	8.1165	8.1297	0.162
16	82	8.1231	8.1297	0.081

Motor field checks. For each of the four geometries, MAPDL advanced a frozen 30 s state through ten increments spanning 0.05 s. Across all four cases the largest nodal difference from a separate brick assembly was 8.16×10^{-6} K. These comparisons test heat storage and transport with the material state and loads held fixed over that interval.

First CF liner-loss event. For the sustainer CF, MAPDL consumed the last 0.453 mm of char in 9.178 s on 20 radial cells. The starting temperature and the casing temperature history came from the independent model at and after 95.08 s of continuous exposure. The maximum nodal difference from the matched calculation was 0.0044 K. Section 18 gives the separate full-exposure checks.

Separate pulse control. In the final 320-row CF control, heating ended at 7 s. The casing continued to warm during cooling and first exceeded 150 °C between 29.4375 and 29.4687 s. At that point 4.718 mm of the original 4.7625 mm liner remained. All 5,295 accepted steps passed the balance checks. The largest sampled native versus radial temperature difference was 0.0151 K.

16 How long the final char layer lasts in the CF cases

At the first saved state where the modeled casing reaches 1000 °C, the independent calculation supplies the residual liner thickness, the radial temperature profile and the subsequent temperature history at the casing. MAPDL then solves heat conduction and surface recession until the char thickness reaches zero. Less than six parts per million of the starting liner mass in either window is still virgin material. We treat this residue as pure char. If exhaustion takes longer than the supplied history, the interface holds its last supplied temperature through the final fraction of the window.

CF case	Handoff [s]	Char left [mm]	Native duration from handoff [s] 20 / 0.010	40 / 0.010	40 / 0.005
Sustainer	95.08	0.453	9.17841	9.17828	9.17741
Booster	129.70	0.404	8.25136	8.25121	8.25033

The three column labels give radial cells and the maximum step in seconds. Each run shortens its last step to the time needed to consume the remaining char.

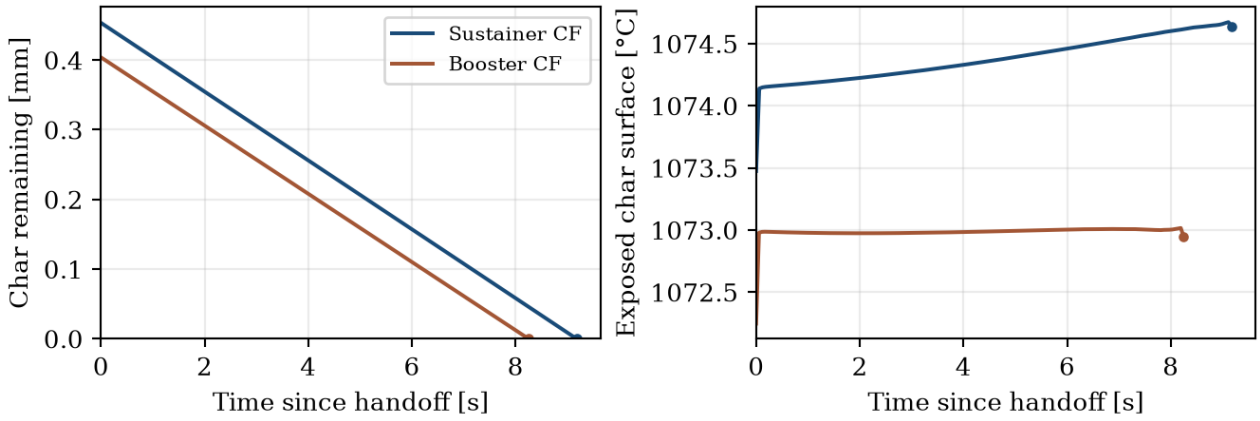


Figure 9: Finest native histories under the prescribed interface temperature. Dots mark exhaustion at the end of each window.

CF case	Refinement	Event shift [ms]	Max. thickness change [μm]	Max. hot face change [K]
Sustainer	Mesh	-0.138	0.007	0.482
Sustainer	Time step	-0.866	0.045	0.437
Booster	Mesh	-0.144	0.007	0.488
Booster	Time step	-0.882	0.046	0.484

Mesh refinement doubles the cell count at a fixed 0.010 s step ceiling. Time refinement halves that ceiling at a fixed 40 cells. A positive event shift means the finer run reaches exhaustion later. For thickness and temperature we compare every saved step of both runs over their shared duration and linearly interpolate the other run at that time. The final-step time intervals overlap for every pair.

The largest nodal discrepancy across the six runs against matched radial heat solutions was 0.0052 K. Summed absolute step energy discrepancies stayed below 0.006% of the boundary energy exchanged in each run. The native windows verify final char consumption conditional on the prescribed casing temperature. A physical firing time also depends on the heat load and casing response above the temperature covered by the CF properties.

17 Native checks with an evolving casing

Here MAPDL solves temperatures through both the liner and the casing. At the interface the liner and casing share one temperature degree of freedom. The controller advances the aluminium fusion enthalpy or CF decomposition after each native field solve. The hot surface moves according to the remaining solid inventory and the calculated mass flux. Two cold-start pilots also advanced virgin sustainer walls from ambient temperature through their first 0.1 s. Their largest difference from independently assembled nodal temperatures was 5.7×10^{-11} K.

Handoff near exhaustion. The independent model supplied a complete state about 0.2 s before its predicted event for each motor. From that state MAPDL solved both sides of the interface until the last char disappeared. The casing was already fully liquid in the aluminium models or above the CF property range at handoff. These tests verify the last few coupled steps. The elapsed time before handoff belongs to the independent model.

Motor	Handoff [s]	Liner at handoff [μm]	Native solves	Native interval to loss [s]	Max. edge ratio†
Sustainer Al	110.40	6.97	5	0.17596	13.64
Sustainer CF	104.16	8.50	5	0.17157	13.64
Booster Al	147.58	4.73	5	0.16558	18.26
Booster CF	137.86	8.88	6	0.18124	12.73

†Largest brick edge divided by its shortest edge, screened over the independent full history using the revised geometry. All four native terminal checks finished without a solver warning or error. Across their archived fields the largest nodal difference from the matched radial system was 2.3×10^{-9} K.

Why change the final step? With the original sector the final brick edge ratios ranged from 176 to 4065 as the wall became extremely thin. The revised sector spans 0.03°. It has two cells circumferentially and two axially, with axial pitch equal to the circumferential arc at the liner–casing interface. The radial mesh starts with 80 cells through the liner and 20 through the case. With one liner cell left, the controller estimates the time to consume the remaining material at the current rate. It shortens the penultimate step when needed so that the final increment retains enough physical thickness to mesh. The following solve recomputes the mass flux, available char and event time. In each independent exposure this adjustment occurred once, with a maximum shift of 0.00027 s in exhaustion.

```
"""Preserve a well-resolved final interval without exceeding the time-step ceiling."""


def end_step(wall,state,requested,ceiling):
    if len(state.nodes)==2:
        radius, end_dt, _, _ = wall.motion(state,2*ceiling)
        # Avoid spending almost an entire ordinary step then leaving a thin
        # sliver to solve. Reserve most of one normal step for exhaustion.
        if radius==wall.interface and requested<end_dt<requested+.90*ceiling and end_dt>.90*ceiling:
            return end_dt-.90*ceiling
    return requested
```

18 Native ANSYS from first exposure to burnthrough

The liner and casing start at 22 °C. From first exposure to liner loss, the model holds the gas at 2504 K and 800 psi with a nominal film coefficient of 1000 W/(m^2 K). After each MAPDL thermal solve the controller updates chemistry, surface position and casing state.

Motor	Casing event [s]	Liner left [mm]	ANSYS gone [s]	Radial gone [s]	Difference [s]	Interval gap [s]
Sustainer Al	99.14	0.430	110.58	110.47	+0.102	0.066
Sustainer CF	95.16	0.453	104.33	104.25	+0.078	0.042
Booster Al	139.44	0.229	147.75	147.61	+0.136	0.100
Booster CF	129.78	0.404	138.04	137.95	+0.095	0.059

The casing event marks complete Al fusion or the CF reaching 1000 °C. The liner thickness there is interpolated between half-second samples. Times use the upper ends of the exhaustion intervals. The gap separates their nearest ends. Beyond the casing event, the thermal model holds the molten Al or CF residue in place. Casing pressure capacity remains undetermined.

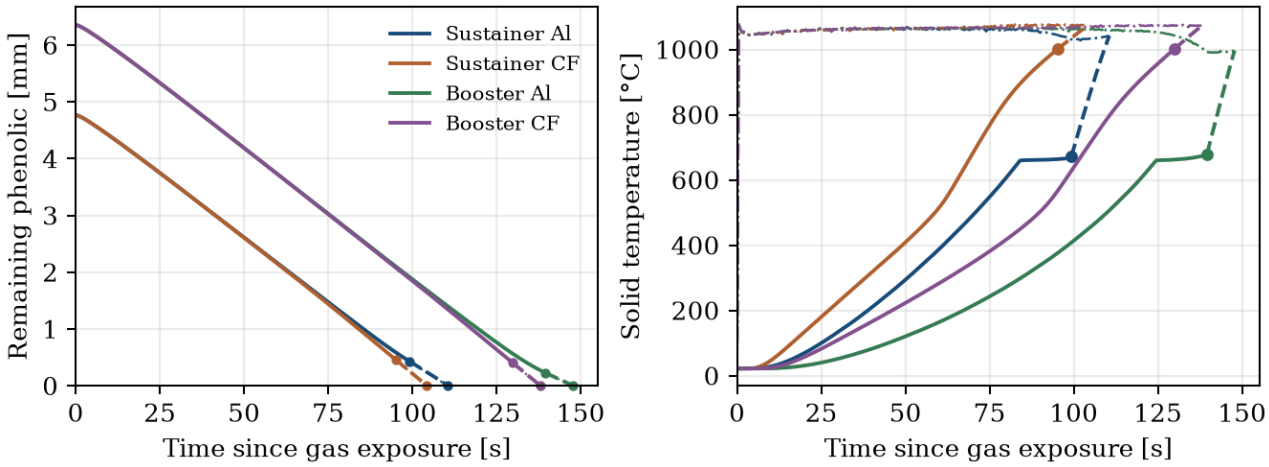


Figure 10: Left, native liner thickness and independent finite-volume thickness (dotted). Right, native interface and exposed liner (dash-dot) temperatures. The final surface point precedes liner loss. Dashed lines use the retained casing model after complete fusion or the CF property limit.

Motor	Native field solves	Max. thickness diff. [μm]	Film dose [MJ/m^2]	Peak live face [°C]	Time and thickness
Sustainer Al	2,903	4.03	131.91	1076	Pass
Sustainer CF	2,713	3.86	122.86	1077	Pass
Booster Al	3,883	4.33	174.82	1076	Pass
Booster CF	3,590	4.64	160.83	1076	Pass

Thickness differences use common half-second samples while liner remains. Film dose is heat entering the moving wall per original exposed area. A pass requires event times within 1% and thickness within the larger of 0.02 mm and 1% of initial thickness. Both solvers use the same heating and char law.

Replaying all native fields with the sealed controller and an independent radial thermal system gave a maximum discrepancy of 2.0×10^{-9} K. The largest absolute energy imbalance was 8.1×10^{-10} of the film dose.

19 How strongly does the char-removal assumption affect time?

The local surface law charges an effective energy L_c for each kilogram of char removed. The earlier energy inventory shows why this coefficient matters. To vary it in isolation, we keep the gas at 2504 K and 800 psi, the nominal film coefficient at 1000 W/(m^2 K), and the same kinetics, char yield, case geometry and casing response. We solve again with half and twice the inherited 35 MJ/kg. These values bracket an assumption. No Iris sample measurement establishes a probability distribution or physical bounds.

	Independent time from exposure to liner loss [s]		
Motor	17.5 MJ/kg	35 MJ/kg	70 MJ/kg
Sustainer Al	71.8	110.5	184.7
Sustainer CF	64.9	104.3	179.0
Booster Al	95.9	147.6	246.7
Booster CF	85.4	137.9	237.7

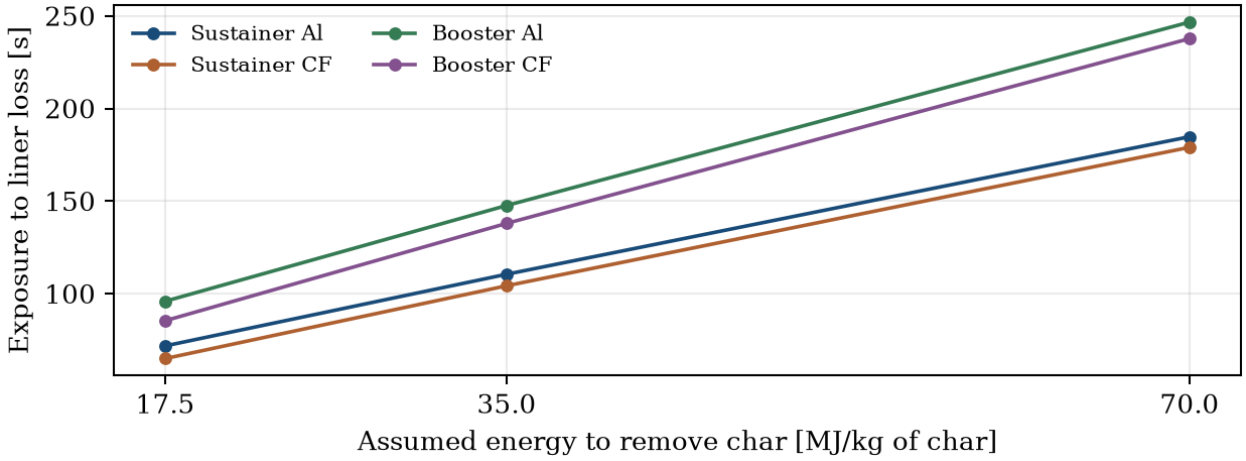


Figure 11: The symbols are separate finite-volume simulations. Lines join the sampled settings. The Al and CF backings evolve during every case.

An increased removal cost leaves less of the incoming heat available to consume char at the surface. Temperature, heat transfer from the gas, and the available char also change during each run. The time therefore need not scale in direct proportion to L_c. The time-step and radial mesh checks use separate refinements. Each published value passes the time, thickness and mass/energy checks, and has a separate nodal finite-element comparison.

A useful calibration would measure the applied heat flux, time-resolved surface recession and remaining char for a specimen of the actual liner. A residual thickness alone cannot separate excessive heating from an underestimated removal cost. The film study varies the heat load separately.

20 How precisely is liner exhaustion resolved?

We repeated each continuous exposure eight times using the independently assembled radial nodal system from the sealed sector controller. All 32 runs started cold and advanced to zero phenolic mass. At each refinement we held the gas load, geometry, chemistry and evolving casing model fixed. The casing mesh always had one quarter as many radial cells as the liner.

Motor	Coarse nodal [s]	Fine nodal [s]	Reference FV [s]	Difference [ms]	Gap [ms]
Sustainer Al	110.576	110.482	110.474	+8.6	6.4
Sustainer CF	104.332	104.257	104.254	+3.5	1.2
Booster Al	147.746	147.623	147.610	+13.6	11.3
Booster CF	138.041	137.951	137.946	+4.1	1.8

The coarse column uses the native production settings (80 liner cells, 40 ms maximum step). The fine column uses 640 cells and 2.5 ms. The reference finite-volume (FV) times use the previously published 640-cell calculation with a 20 ms maximum step. Each time is the upper end of the final integration interval. The gap measures the separation between the nearest ends of the two final intervals. Decimal places here expose differences between calculations.

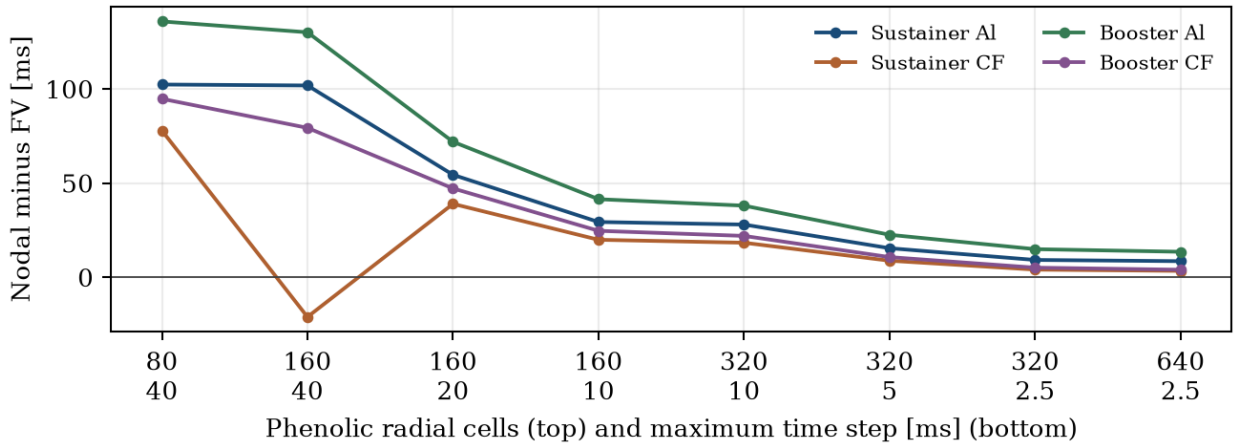


Figure 12: Change in predicted exhaustion time relative to the same FV reference for each motor. Adjacent settings change either space or time resolution. The temporary dip for the CF sustainer occurs at the coarse 40 ms step.

At 320 liner cells, halving the maximum step from 5 to 2.5 ms changes the endpoint by at most 7.6 ms. At 2.5 ms, increasing from 320 to 640 cells changes it by at most 1.4 ms. The fine nodal and reference FV estimates support the same whole-second scale. The positive gaps still matter for a claim of tighter agreement. Section 21 repeats the comparison at matched mesh and time settings to separate spatial and temporal effects. Material and firing assumptions limit the physical interpretation throughout.

21 Final time intervals at matched resolution

Both independent methods use the same number of radial cells and the same maximum time step in each comparison. We first halve the step at 320 liner cells (5 to 2.5 ms), then double the liner mesh to 640 cells at 2.5 ms. The casing has one quarter the liner cell count. All runs impose the same continuous nominal heating and evolve the casing from ambient temperature.

Motor	Fine FV [s]	Fine nodal [s]	FV time-step change [ms]	FV mesh change [ms]	Final gap [ms]
Sustainer Al	110.4763	110.4822	+0.35	-2.10	3.62
Sustainer CF	104.2531	104.2571	-0.11	-1.93	1.71
Booster Al	147.6159	147.6233	+0.86	-3.85	5.10
Booster CF	137.9460	137.9505	-0.10	-2.84	2.28

The time columns give the upper ends of the last integration intervals. Each change column subtracts the previous finite-volume endpoint while varying just the indicated setting. At the finest common resolution, the largest difference between method endpoints is 7.35 ms. At common half-second stations, the largest thickness difference is 0.23 μm and the largest casing-interface temperature difference is 0.17 K.

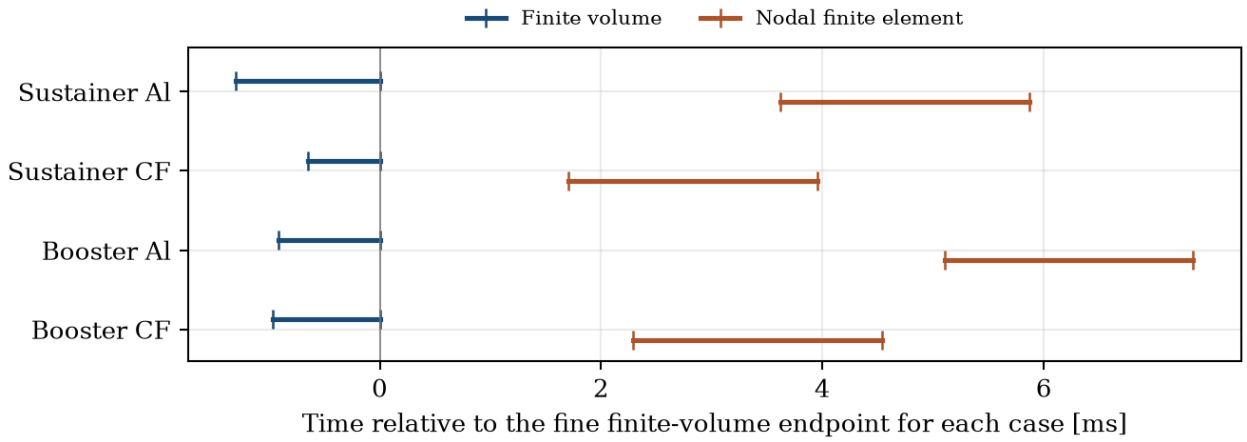


Figure 13: Each segment runs from the last saved step with liner present to the numerical exhaustion event. The origin in each row is its own FV endpoint. The clear space between segments is the gap reported in the table.

The final numerical intervals are separated in all four cases. Their width reflects the integration step alone. The unknown firing heat flux, recession chemistry and post-limit casing response introduce separate physical uncertainties.

Output-time recovery. One FV run approached the 142 s sample to about 1.0×10^{-10} s and attempted a separate step of that duration. Its energy check rejected the step. The successor splits the preceding interval into two comparable steps whenever the next sample would otherwise leave a remainder below 5% of a normal step. It resumed the Al booster from its saved 126.235 s state, retaining its mass and heat histories. The seven already completed FV paths kept their original records. All twelve comparisons use the original balance tolerances.

22 Comparison with the preceding finite-row model

For a continuously exposed liner, the inherited independent model reaches the casing after roughly two minutes for the sustainer and two and a half minutes for the booster. Each event occurs after the casing reaches the limit of its property evidence. Actual firing heat flux, local exposure duration and measured liner loss are unknown. The times below describe a hypothetical intact backing under sustained heating. Physical motor burnthrough remains open.

	Time to final liner-row loss [s]			Property flag [s]
Case	160, full step	160, half step	320, half step	320, half step
Sustainer Al	114.95	114.21	115.10	49.49
Sustainer CF	112.18	111.42	112.30	18.34
Booster Al	151.70	151.03	151.95	87.27
Booster CF	147.05	146.36	147.27	32.71

Table 3: Independent radial calculations. The initial thicknesses are 4.7625 mm for the sustainer and 6.3500 mm for the booster. A flag marks the first 150 °C CF or 300 °C Al interface crossing. The table separates this evidence limit from complete liner-row loss.

The gas stays at 2504 K (2230.85 °C), with nominal pressure 800 psi and uncorrected film coefficient 1000 W/(m^2 K). Outgassing modifies convection. Effective hot-side emissivity is zero. The outside sees 22 °C, a film coefficient of 8 W/(m^2 K) and emissivity 0.25. The hot surface temperature evolves as the wall absorbs heat. The local liner face is exposed from time zero until exhaustion. Sustained heating is a chosen scenario.

The three columns use 160/40, 160/40 and 320/80 radial cells in the liner and casing. The full step ceiling is 0.0015625 s and the half step is 0.00078125 s. In all twelve records the final row disappears within that run’s time increment. Resolution changes move the event by more than an individual time increment, so digits beyond those shown carry little meaning.

23 Liner remaining and casing temperature

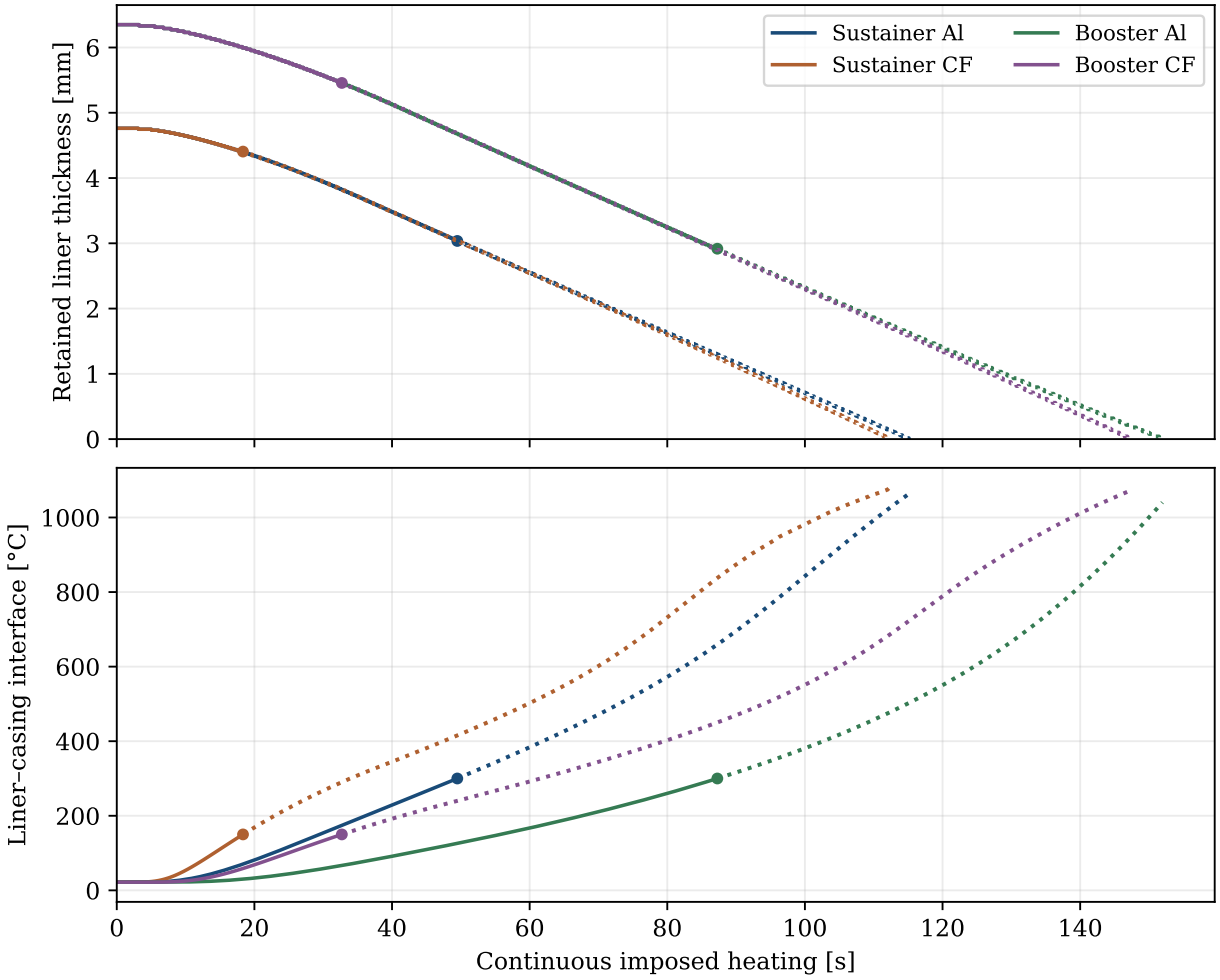


Figure 14: Independent 320/80 finite-row results. Dots mark the first casing-property flag. The dotted continuation assumes an intact casing with the inherited surrogate properties. Thickness includes retained char and is distinct from the thickness of virgin material. Similar curves overlap near ignition.

Al uses the inherited constant properties. The CF backing uses its radial conductivity and a heat-capacity correlation clamped to the inherited interval. The plotted temperatures can exceed the materials' supported range by hundreds of degrees. Integrity of a hot pressurized case, resin decomposition, bond failure and local cracks require additional evidence. The previous 7 s pulse and cooling results, native ANSYS verification and complete MAPDL and Fortran listings are in the companion native report. Earlier rejected endurance jobs are excluded from every comparison here.

24 What changes with time step and mesh?

Time comparisons halve the time step at fixed 160/40 mesh. Mesh comparisons double the number of cells in both materials at the same half step. All other physics inputs match between each pair. A signed time change is the finer result minus the coarser result.

	Event change [s]		Remaining thickness [mm]		
Case	Time	Mesh	Time diff.	Mesh diff.	Target
Sustainer Al	-0.74	+0.89	0.0595	0.0595	0.0476
Sustainer CF	-0.76	+0.88	0.0595	0.0595	0.0476
Booster Al	-0.67	+0.93	0.0397	0.0595	0.0635
Booster CF	-0.69	+0.91	0.0397	0.0595	0.0635

Table 4: Thickness columns give the largest absolute difference on a shared 0.5 s grid up to the first exhaustion. Every row transition was archived, so sampling uses the retained row count at each comparison time.

The largest change in event time is 0.78 percent of the finer calculation. 4 of eight comparisons meet the intermediate-thickness target and 0 of eight pairs have overlapping final-event brackets. Both sustainer pairs miss the 0.047625 mm target. Both booster pairs meet their 0.0635 mm target on the shared grid and over the full common history. Halving the step advances the event in all four cases, while doubling the mesh count delays it. Comparing only the first and third columns would partially cancel those effects. Each row retains its full thermal capacity during partial consumption until its residual-mass criterion removes it. The moving-wall treatment instead tracks the face continuously.

A separate thin-coupon test paused the independent worker after the second row loss and casing flag. A fresh controller then recovered the same complete history, final-event bracket and byte-identical terminal checkpoint as uninterrupted integration. The audit reconstructs final mass from the saved states, checks each row transition, and matches source and input fingerprints across all twelve records. Maximum cumulative relative mass error is 1.54e-14. Maximum relative residual of the inherited discrete step energy equation is 9.22e-11. The new material and wall equations below use a separate open-system energy ledger. The 585-node native conduction results and moving-wall benchmarks appear in Section 15. Section 18 gives the accepted full-exposure MAPDL results. Measured heating and recession, high-temperature backing data and sensitivity to surface chemistry remain necessary for the physical interpretation.

25 A moving surface with separate virgin, char and gas inventories

The candidate transports the masses of virgin phenolic M_v, char M_c, and their combined sensible enthalpy H_s in each cylindrical cell. A cell has volume $(r_o^2 - r_i^2)/2$ per radian per metre of axial length. Conservative overlap integration in $r^2/2$ returns both the retained and swept inventories as the face advances. In the equations below, M and H may refer either to a physical cell or to these quantities per radian per metre, provided every source uses the same convention.

25.1 Bulk reaction and reconstruction

We use a common reference $T_* = 22\,^\circ\mathrm{C}$ for virgin solid, char and gas. The inherited table defines piecewise-linear specific heat capacities. Let $h_i(T) = \int_{T_*}^{T} c_{p,i}(u)\,du$ for $i = v, c$ and $h_g(T) = 1600(T - T_*)$ J/kg when T is in degrees Celsius. The candidate holds the first heat capacity constant between 0 and 22 degrees Celsius and rejects states above the final 2476.85 degree table entry or below zero. Positive phase masses give a strictly increasing enthalpy relation $H_s = M_v h_v(T) + M_c h_c(T)$. Inverting the quadratic integral within its table interval recovers a single temperature after remapping. An exhausted cell carries zero mass and enthalpy and has no temperature to reconstruct.

Let $y = 600/1250 = 0.48$ denote the assumed char yield per kilogram of virgin material consumed. With $\bar{T} = (T^n + T^{n+1})/2$ and $k = 333\exp[-64081/(R(\bar{T} + 273.15))]$ in reciprocal seconds, the update uses

$$d = M_v^n(1 - e^{-k\Delta t}), \quad M_v^{n+1} = M_v^n - d, \quad M_c^{n+1} = M_c^n + yd, \quad m_g = (1 - y)d.$$

Mass and enthalpy use the same d. The local scalar temperature solve, and the converged coupled cell equation, enforce

$$H_s^{n+1} - H_s^n = Q_{\mathrm{cell}} - L_p d - m_g h_g(\bar{T}).$$

The chosen basis for $L_p = 418$ kJ/kg is virgin mass consumed at T_*. Gas leaves carrying its sensible enthalpy at generation. The phase inventories on the left already include the change in stored solid sensible heat.

25.2 Partially converted material at the wall

Let a and b be the virgin and char mass densities in the material adjacent to the gas. The current candidate assumes that residual virgin material swept by the front completes pyrolysis there and its char reacts at that same face. For speed $v \geq 0$ the outgoing surface pyrolysis gas flux is $j_g = (1 - y)av$ and the outgoing char-product flux is $j_c = (b + ya)v$, in kg/(m^2 s). Their sum equals the rate $(a + b)v$ at which the surface sweeps up solid. Writing $\alpha = b/(b + ya)$, we impose $j_c = \alpha j_{\mathrm{law}}(T_s, p)$ with the inherited kinetic and mass-transfer law. A virgin wall can heat and pyrolyse volumetrically before surface recession begins.

For a first-cell temperature T_1, define the sensible power delivered to the boundary by swept inventory as $q_{\mathrm{swept}} = v[ah_v(T_1) + bh_c(T_1)]$. Per unit hot-face area the wall balance is

$$q_{\mathrm{conv}} + q_{\mathrm{rad}} + q_{\mathrm{swept}} = q_{\mathrm{cond}} + j_g h_g(T_s) + j_c h_c(T_s) + avL_p + j_c L_c,$$

$$q_{\mathrm{cond}} = \frac{\displaystyle\int_{T_1}^{T_s} [(1 - \alpha)k_v(T) + \alpha k_c(T)]\,dT}{r_s \log(r_1/r_s)}.$$

Here $q_{\mathrm{cond}} > 0$ flows into the retained solid, r_s is the moving wall radius and r_1 the adjacent cell centre. The conduction expression integrates the cylindrical resistance at fixed local composition. The film coefficient uses total outward mass flux, including bulk gas arriving at the surface and both j_g and j_c. A bracketed scalar solve closes film heating, optional radiation, conduction and recession at the same T_s. Bulk gas has already exported its sensible heat in the bulk equation. The chosen proxy $h_c(T_s)$ for the char product and excess cost $L_c = 35$ MJ/kg together specify a provisional removal energy. Data for reaction products, recession and pyrolysis are needed to determine whether these assumptions hold for the Iris liner.

25.3 Verification to date

Tests compare the enthalpy primitive against independent quadrature and the isothermal reaction against the analytic solution. In a four-second manufactured heated cell, doubling the time steps from 8 to 16, 32 and 64 reduces temperature error against an independent ODE integration from 0.151 to 0.0376, 0.00940 and 0.00235 K. A nonuniform reacting shell with prescribed recession closes the integrated mass and energy ledgers through final-cell removal to relative errors below 2×10^{-15}. Separate manufactured tests recover imposed face temperatures for a virgin conducting wall, pure reacting char, and a mixed wall with blowing and radiation. The face equation residual is below 3×10^{-9} W/m^2 for those fixtures. These checks exercise individual material and boundary balances. The opening sections give the coupled transient checks and event comparisons. Native verification of this formulation remains open.

25.4 Radial heat flow between cells

On a stationary mesh the conduction operator transfers heat between adjacent cell centres through the two cylindrical half-cell resistances. For centre radii r_i, r_{i+1} and shared face r_f, the heat flow per radian per axial metre toward increasing radius is

$$F_{i+1/2} = \frac{T_i - T_{i+1}}{\log(r_f/r_i)/k_i + \log(r_{i+1}/r_f)/k_{i+1}}.$$

Backward Euler balances each cell's change in sensible heat against the difference of its two face flows and any supplied power. A two-layer steady wall reproduces the logarithmic temperature solution to within 6×10^{-13} K. A transient annular conduction test compares against an analytic Bessel eigenmode over one decay time. With the time error removed to first order, the maximum errors on 16, 32 and 64 cells are 0.118, 0.0295 and 0.00738 K. These tests exercise diffusion and the material interface at fixed geometry. The coupled checks extend these tests to simultaneously evolving temperature, gas flow and surface position.

Reference for the method. Chen and Milos describe a moving-grid treatment of decomposition, energy and surface balances for a charring ablator. Their PICA data apply to a different material and environment. NASA NTRS 20130014432.

A Conservative geometry and remapping code

Below is the complete independent geometry kernel. The accompanying model directory contains the bulk reaction, boundary and conduction implementations and their verification programs.

```
"""Conservative radial geometry kernel for a future moving-surface model.

This is NOT a heat-transfer/reaction solver or a recession law. It transports
extensive states under a prescribed boundary movement. Units are per radian
and per metre of axial length; multiply by sector angle and length if needed.
"""
from dataclasses import dataclass
import numpy as np


@dataclass
class Remap:
    density: np.ndarray
    cell_integrals: np.ndarray
    removed_integrals: np.ndarray
    original_integrals: np.ndarray
    balance_residual: np.ndarray


def advance_front(radius, interface, speed, dt):
    """Locate geometric exhaustion for a prescribed constant normal speed.

    The caller must obtain speed from a verified surface mass/energy balance.
    A temperature, conversion threshold or arbitrary cutoff is not used here.
    """
    if not np.all(np.isfinite([radius, interface, speed, dt])):
        raise ValueError('Nonfinite front inputs')
    if radius <= 0 or interface < radius or speed < 0 or dt <= 0:
        raise ValueError('Invalid front inputs')
    if radius == interface:
        return interface, 0., True
    if speed == 0:
        return radius, dt, False
    to_event = (interface - radius) / speed
    if to_event <= dt:
        return interface, to_event, True
    return radius + speed * dt, dt, False


def remap(old_edges, density, new_edges):
    """Integrate piecewise-constant conserved densities over new radial cells.

    The last dimension of density is the old cell index. Other dimensions can
    hold virgin-equivalent mass, char mass, solid enthalpy and related conserved
    quantities. Remap those quantities, not temperature or alpha independently.
    Thermodynamic reconstruction and reaction/removal source terms are the
    caller's responsibility. Interface positions must be retained by the caller
    so heterogeneous material labels are not averaged into one material.

    New cells cover the retained domain [new inner radius, old outer radius].
    A one-entry new_edges array denotes exact exhaustion of this domain. The
    swept-out mass and enthalpy are returned explicitly, not silently deleted.
    """
    old = np.asarray(old_edges, dtype=float)
    new = np.asarray(new_edges, dtype=float)
    field = np.asarray(density, dtype=float)
    if old.ndim != 1 or new.ndim != 1 or len(old) < 2 or len(new) < 1:
        raise ValueError('Expected radial edge arrays')
    if field.ndim < 1 or field.shape[-1] != len(old) - 1:
        raise ValueError('Density field does not match old cells')
    if not all(np.all(np.isfinite(x)) for x in [old, new, field]):
        raise ValueError('Nonfinite remap inputs')
    if old[0] <= 0 or np.any(np.diff(old) <= 0) or np.any(np.diff(new) <= 0):
        raise ValueError('Cells must have strictly positive radial volume')
```

```
    if new[0] < old[0] or new[-1] != old[-1]:
        raise ValueError('Only inner recession with a fixed outer boundary is supported')
    # x = r^2/2 is the exact cylindrical volume coordinate per radian/metre.
    xold = old * old / 2
    xnew = new * new / 2
    old_integrals = field * np.diff(xold)
    cumulative = np.concatenate([np.zeros(field.shape[:-1] + (1,)),
                                 np.cumsum(old_integrals, axis=-1)], axis=-1)
    flat = cumulative.reshape((-1, len(old)))
    retained_cumulative = np.array([np.interp(xnew, xold, row) for row in flat])
    retained_cumulative = retained_cumulative.reshape(field.shape[:-1] + (len(new),))
    integrals = np.diff(retained_cumulative, axis=-1)
    removed = retained_cumulative[..., 0]
    original = cumulative[..., -1]
    residual = original - removed - integrals.sum(axis=-1)
    new_density = integrals / np.diff(xnew)
    return Remap(new_density, integrals, removed, original, residual)
```

B Native continuous-exposure inputs

The complete entry deck for the aluminium pilot below checks the controller and configuration before transferring control to the solver's embedded Python interpreter. The four complete exposures use the same controller archive. Each macro-step generates a conventional MAPDL input file, solves one transient thermal field and exports every node for replay. The previous pulse study's full Fortran material routine and MAPDL listings are printed in the companion native report. Here the Python controller advances chemistry and recession between finite-element thermal steps.

```
/CLEAR,NOSTART
! In-process controller. Native execution and acceptance are separate.
*PYTHON
import hashlib, json, sys
from pathlib import Path
pkg = Path('\\\\Mac\\Home\\Developer\\MIT-RT-26_27-Work\\simulation_work\\solidprop\\iris\\v1\\
    configs\\native_verification\\whole_sector_r2')
archive = pkg / 'controller.zip'
signature = '6ff2c7ff92e9fde4eaa3fa8a21032247a684c12bf2306a6fde5ecb00c653abf8'
if hashlib.sha256(archive.read_bytes()).hexdigest() != signature:
    raise RuntimeError('Controller package hash mismatch')
raw = (pkg / 'start2al_settings.json').read_bytes()
if hashlib.sha256(raw).hexdigest() != '
    ff6b58ff77877bffa0b7afeec565ca8eb6625a27a4d4ffa16cbcfaebc86928cb':
    raise RuntimeError('Configuration hash mismatch')
sys.path.insert(0,str(archive))
from sector_mapdl import execute
execute(mapdl,json.loads(raw),signature,'\\\\Mac\\Home\\Developer\\MIT-RT-26_27-Work\\simulation_work
    \\solidprop\\iris\\v1')
*ENDPY
/COM,IRIS_WHOLE_SECTOR_RETURNED
/EXIT,NOSAVE
```

Sector weights used by the command generator.

```
def topology(block, geometry):
    nt, ny = geometry['angular_divisions'], geometry['axial_divisions']
    theta = math.radians(geometry['angle_deg'])/nt
    dy = geometry['length_m']/ny
    # Shape-function weights around one ring. End nodes belong to half as many bricks.
    wtheta = np.ones(nt+1)
    waxial = np.ones(ny+1)
    wtheta[[0,-1]] = waxial[[0,-1]] = .5
    weights = np.outer(wtheta,waxial).ravel()/(nt*ny)
    scale = nt*math.sin(theta)*geometry['length_m']
    if not math.isclose(block.facet, math.cos(theta/2), abs_tol=1e-14):
        raise ValueError('Inconsistent facet geometry')
    return nt, ny, theta, dy, weights, scale
```

The ANSYS command and element references document the in-process Python interface, SOLID278 heat and film options, and MASS71 thermal capacitance.

Complete generator for one thermal step

```
def commands(block, geometry, sequence):
    nt, ny, theta, dy, weights, scale = topology(block, geometry)
    width = len(weights)
    r = block.radii
    nr = len(r)-1
    count = nr*nt*ny
    extra = block.capacity-volume(dual(r))
    if np.min(extra)<=0:
        raise ValueError('Specified thermal mass is below preliminary unit volumetric capacity')
    lines = ['/NOPR','/PREP7']
    if sequence>1:
        lines += ['ALLSEL,ALL','FDELE,ALL,ALL','EDELE,ALL','NDELE,ALL']
    else:
        lines += ['ET,1,SOLID278','KEYOPT,1,13,1','KEYOPT,1,15,1','ET,2,MASS71','KEYOPT,2,3,1']
    def nid(i,j,k):
        return 1+i*width+j*(ny+1)+k
    for i, radius in enumerate(r):
        for j in range(nt+1):
            angle=(j-.5*nt)*theta
            for k in range(ny+1):
                node=nid(i,j,k)
                lines.append(f'N,{node},{radius*math.sin(angle):.16g},{k*dy:.16g},{radius*math.cos(
                    angle):.16g}')
    lines += ['TYPE,1','REAL,1']
    for i, conductivity in enumerate(block.k):
        mat = i+1
        lines += [f'MP,KXX,{mat},{conductivity:.16g}',f'MP,DENS,{mat},1',f'MP,C,{mat},1',f'MAT,{mat}'
            ]
        for j in range(nt):
            for k in range(ny):
                corners = [nid(i,j,k),nid(i+1,j,k),nid(i+1,j+1,k),nid(i,j+1,k)]
                points = corners+[a+1 for a in corners]
                lines.append('EN,'+','.join(map(str,[1+(i*nt+j)*ny+k,*points])))
    lines += ['TYPE,2','MAT,1']
    for i, cap in enumerate(extra):
        for ring_id, weight in enumerate(weights):
            node=1+i*width+ring_id
            lines += [f'R,{node},{cap*scale*weight:.16g}',f'REAL,{node}',f'EN,{count+node},{node}']
    lines += ['FINISH','/SOLU','ANTYPE,TRANS,NEW','TRNOPT,FULL','TINTP,,,,1','KBC,1','AUTOTS,OFF',
              'NSUBST,1,1,1',f'TIME,{block.dt:.16g}']
    for i, initial in enumerate(block.initial):
        for ring_id, weight in enumerate(weights):
            node=1+i*width+ring_id
            lines.append(f'IC,{node},TEMP,{initial:.16g}')
            if block.load[i]:
                lines.append(f'F,{node},HEAT,{block.load[i]*scale*weight:.16g}')
    for i, face, conv, radius in [(0,5,block.hot_h,r[0]),(nr-1,3,block.outer_h,r[-1])]:
        # Bulk-temperature terms are already in nodal load. This adds the
        # matching film matrix once, against a zero-Celsius reference.
        for e in range(1+i*nt*ny,1+(i+1)*nt*ny):
            lines += [f'SFE,{e},{face},CONV,0,{conv*block.facet/radius:.16g}',f'SFE,{e},{face},CONV
                ,2,0']
    lines += ['OUTRES,ALL,LAST','SOLVE','FINISH','/POST1','SET,LAST','*GET,IRST,ACTIVE,0,SET,TIME',
              '*CFOPEN,sector_field,txt',f'IRSEQ={sequence}','*VWRITE,IRSEQ,IRST','(F12.0,1X,E25.16)'
                  ,
              f'*DO,IRNODE,1,{width*len(r)}','*GET,IRTEMP,NODE,IRNODE,TEMP','*VWRITE,IRNODE,IRTEMP',
              '(F12.0,1X,E25.16)','*ENDDO','*CFCLOSE','FINISH','/GOPR']
    return '\n'.join(lines)+'\n'
```